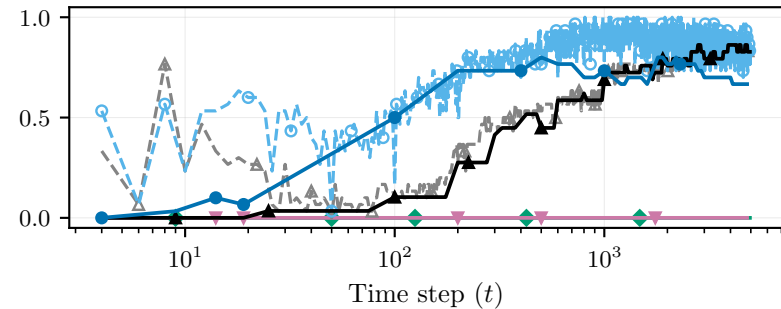


Confidence $1 - \hat{\beta}_t$ 

Online clustering dynamic regret

